

# ChatGPT and academic integrity: the role of personal best goals, academic competence, and workplace stress

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## Abstract

**Purpose** – The research aims to unravel the dynamics of academic integrity in the ChatGPT era by analyzing critical predictors such as personal best goals (PBG), academic competence and workplace stress. Furthermore, it examines how ChatGPT adoption acts as a moderating factor, potentially influencing the relationship between these predictors and academic integrity, offering a nuanced understanding of its impact in academic settings.

**Design/methodology/approach** – Relying on the social cognitive theory, the authors adopt a quantitative approach through an online survey to explore the impact of key variables – PBG, academic competence and workplace stress, as well as ChatGPT adoption on integrity – by analyzing data from responses collected through Academic Social Networking Sites.

**Findings** – The study found that PBG related positively to academic integrity among academic staff. However, workplace stress had a negative impact on academic integrity, while academic competence failed to report any effect. With the integration of ChatGPT adoption as a moderator into the model, the authors found the association between PBG and academic integrity altered to be negative. The ChatGPT adoption-moderated interactions of academic competence and workplace stress on academic integrity were significant.

**Practical implications** – The findings suggest that institutions should provide training on the ethical and effective use of artificial intelligence (AI) tools like ChatGPT to ensure they support rather than hinder academic integrity. Additionally, organizations should focus on stress management initiatives and fostering a balanced approach to personal goal setting to mitigate the potential negative impacts of ChatGPT adoption on ethical behavior.

**Originality/value** – This attempt is both timely and pioneering, addressing a clear gap in the current body of literature. As limited studies have examined the role of ChatGPT in academic settings, this work stands out as one of the earliest to investigate how educators and researchers incorporate OpenAI tools into their professional practices.

**Keywords** ChatGPT, Personal best goals, Academic integrity, Workplace stress, Artificial Intelligence  
**Paper type** Research paper

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## 1. Introduction

Developed by OpenAI, AI-powered chatbots represent cutting-edge achievements in natural language processing, enabling them to engage in complex conversations, assist with a wide range of tasks and exhibit an advanced understanding and text generation that challenges the boundaries of human-machine interactions (Bouteraa *et al.*, 2024b; Dowling and Lucey, 2023; Bin-Nashwan *et al.*, 2025; Dwivedi *et al.*, 2023). These AI tools have the potential to greatly impact education (Ratten and Jones, 2023; Duong *et al.*, 2023; Lo, 2023; Cotton *et al.*, 2024; Alser and Waisberg, 2023). ChatGPT, in particular, demonstrates flexibility in creating personalized learning experiences and offering instant access to

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extensive information, potentially revolutionizing traditional educational practices (Choi *et al.*, 2023). This transformative platform could reshape the educational landscape by enhancing how educators design course materials, develop interactive lessons and engage students in dynamic learning experiences.

Delving into the context of academic integrity, ChatGPT's capabilities have become increasingly significant (Lund and Wang, 2023; Ratten and Jones, 2023), offering both opportunities and challenges within educational institutions (Bin-Nashwan *et al.*, 2023a). By assisting in research and providing innovative teaching materials, ChatGPT can support ethical practices when used responsibly, ensuring proper attribution and adherence to academic standards (Cotton *et al.*, 2024; Bin-Nashwan *et al.*, 2023a). However, its ability to generate content autonomously raises concerns about plagiarism, misuse and the potential erosion of academic honesty if not guided by clear policies and ethical frameworks. Furthermore, while ChatGPT's efficiency in handling routine tasks may allow educators to focus on fostering critical thinking and guiding students through complex topics, it also necessitates an emphasis on teaching students and staff how to use AI tools in a manner that upholds integrity (Meyer *et al.*, 2023; Cotton *et al.*, 2024).

ChatGPT has the potential to reshape the educational landscape (Sadallah *et al.*, 2024; Bouteraa *et al.*, 2024a), but its integration must be accompanied by strategies to safeguard and promote academic integrity in this evolving environment. However, the inclusion of ChatGPT within academia depicts a multifaceted landscape replete with both opportunities and challenges (Lund and Wang, 2023; Rahman *et al.*, 2023). As this technology pervades educational settings, the central concern of reinforcing academic integrity, rigor and ethical conduct becomes a critical focus (Eke, 2023; Bin-Nashwan *et al.*, 2023a). The adoption of AI tools like ChatGPT in research, scholarly pursuits and academic assistance introduces a complex terrain where the boundaries between original thought, collaboration and automated assistance blur (Cotton *et al.*, 2024; Shiri, 2023; Lo, 2023; Alser and Waisberg, 2023). Such challenges necessitate a critical examination of the influence of ChatGPT on the academic landscape, particularly in academic integrity practices.

The present research focuses on the complex relationship between ChatGPT adoption and academic integrity within educational settings. As academia increasingly integrates AI tools, understanding their implications for academic integrity has become crucial. Key issues include the authenticity of scholarly work, the effect of AI-driven chatbots on individual academic competence and stress management and how personal best goals (PBG) align within this AI-centric academic landscape. Existing theoretical discussions on AI adoption in education reveal a significant research gap. Despite the rapid proliferation of tools like ChatGPT in academia (Lo, 2023; Cotton *et al.*, 2024; Alser and Waisberg, 2023; Farrokhnia *et al.*, 2023; Lund and Wang, 2023; Rahman *et al.*, 2023; Dowling and Lucey, 2023; Dwivedi *et al.*, 2023; Jimenez, 2023), empirical studies examining their influence on academic integrity – especially in the context of PBG, academic competence and workplace stress – are notably limited. This lack of in-depth research highlights the importance and timeliness of this study.

## 2. Theoretical framework

### 2.1 Generative artificial intelligence

Generative AI represents a significant technological breakthrough that is reshaping both industry and academia. Unlike traditional AI systems, generative AI is capable of producing new content – such as text, images, music and other media – by learning patterns from vast data sets. Tools like OpenAI's ChatGPT exemplify this category of AI, which is now transforming processes in a variety of industries, including education, healthcare, entertainment and manufacturing. In the context of Industry 4.0 and 5.0, generative AI enhances automation, fosters innovation and promotes human-machine collaboration.

These advancements contribute to more agile and efficient systems, helping businesses to remain competitive in an increasingly digital world. Furthermore, as we move toward Society 5.0, generative AI plays a pivotal role in creating more personalized, intelligent solutions across sectors.

In academia, the adoption of generative AI tools like ChatGPT is revolutionizing how scholars approach knowledge creation, research and teaching. However, this transformation also introduces new challenges, particularly concerning academic integrity. Traditional academic practices, such as originality in research and ethical use of sources, face scrutiny as AI-generated content becomes more prevalent. The core issue lies in how these tools influence academic behaviors, integrity and performance standards. To better understand these dynamics, this study builds on the foundations of generative AI in the industry, while addressing its growing impact on academic institutions. The study “ChatGPT and similar generative artificial intelligence (AI) for smart industry: Function, challenges, and opportunities for Industry 4.0, Industry 5.0, and Society 5.0” by [Rane \(2023\)](#) offers an important reference in exploring the capabilities of generative AI. [Rane \(2023\)](#) detailed how AI technologies, including generative models, are leveraged for innovation in smart industries, improving efficiency, creativity and decision-making. The study also highlighted the ethical challenges associated with generative AI, such as biases in algorithms, transparency and the implications of AI-generated content. These challenges mirror similar issues faced in academia, where the line between human-generated and AI-generated work is becoming increasingly blurred.

Drawing on this and related works, such as [Dwivedi et al. \(2023\)](#) and [Cotton et al. \(2024\)](#), this study situates generative AI within a broader theoretical context, linking its adoption in academia to evolving expectations for ethical behavior, competence and personal performance goals. To further strengthen the theoretical framework of this study, we integrate key concepts from social cognitive theory (SCT), which has been widely used to explore human behaviors in educational settings. SCT emphasizes the interaction between personal factors, behavior and environmental influences. Within the context of academic integrity and AI adoption, generative AI tools like ChatGPT influence these three components. For example, they affect academic self-efficacy, as students and faculty use these tools to augment their research and writing, potentially enhancing their perceived competence. However, this technological shift also poses risks to academic integrity, as the ease of generating content can undermine traditional academic values of originality and authorship.

## 2.2 Social cognitive theory

SCT, a prominent psychological framework, emphasizes the critical role of the social environment in shaping human behavior, particularly in areas like motivation, learning and self-regulation ([Bin-Nashwan and Muneeza, 2023](#)). Widely applied across psychological research, SCT has become increasingly relevant to the study of academic integrity ([Schunk and DiBenedetto, 2020](#)). This discussion is grounded in Bandura's SCT ([Bandura, 1986, 1997, 2001](#)), narrowing its focus to explore the connections between this theoretical framework and academic integrity. According to SCT, human motivation and behavior are strongly influenced by learning processes, with forethought – a mechanism of anticipatory control – serving as a cornerstone. This allows individuals to anticipate the consequences of their actions, including those involving ethical decision-making. Several core factors within SCT contribute to understanding behavior ([Luszczynska and Schwarzer, 2005](#)).

A key component is perceived self-efficacy, which reflects a person's confidence in their ability to execute the actions needed to achieve specific outcomes while maintaining ethical standards. [Bandura \(1986\)](#) highlights competence as an essential element of SCT ([Bembenutty et al., 2016](#)). The theory also underscores the role of personal beliefs, self-perceptions and goal-setting in influencing ethical behavior, while simultaneously

recognizing how environmental factors shape decisions and actions (Bandura, 1989). For instance, individuals may face ethical dilemmas or opportunities for misconduct influenced by their social environment, such as observing peers' behaviors or engaging with innovative technologies (Ratten and Ratten, 2007).

In today's academic landscape, emerging technologies like ChatGPT represent a significant environmental factor (Bin-Nashwan *et al.*, 2023a). While such tools hold great potential for fostering ethical practices, they also present risks of misuse, emphasizing the necessity of developing strong self-regulation and ethical foresight among users. SCT suggests that setting PBG can enhance academic integrity, as these goals encourage individuals to pursue higher ethical standards (Burns *et al.*, 2017, 2018). Additionally, intrinsic factors such as perceived stress significantly affect ethical decision-making. Yin *et al.* (2020) demonstrated that stress can undermine self-regulation, potentially leading to ethical lapses in high-pressure scenarios. Similarly, Resurreccion (2012) explored the interplay between academic integrity and intrinsic motivation, highlighting how internal beliefs influence ethical behavior. Doménech-Betoret *et al.* (2017) also identified competence as a critical factor that supports self-regulation and ethical conduct within the SCT framework.

### 2.3 Research model and hypotheses

Drawn on SCT, the selection of each variable was strategically made to address key factors influencing academic integrity in the context of ChatGPT adoption. PBG were included as they represent individual motivations for achieving academic excellence, which can impact the ethical use of AI tools. Academic competence was chosen to assess how the perceived skill and expertise of faculty and researchers influence their ability to navigate AI tools ethically, aligning with academic integrity standards. Workplace stress was incorporated as it can affect decision-making and ethical behavior, potentially leading to shortcuts or unethical practices when using AI-driven tools. Finally, ChatGPT adoption serves as the central variable, reflecting the extent to which AI tools are used and their moderating effects on the relationships between personal goals, competence and workplace stress. These variables collectively provide a comprehensive understanding of the dynamics at play in the AI-driven academic environment.

*2.3.1 Academic integrity.* Academic integrity is described as “the values, behaviour, and conduct of academics in all aspects of their practice” (Macfarlane *et al.*, 2014). Upholding academic and educational integrity requires a strong commitment to truthfulness, fairness, impartiality, respect and responsibility (Holden *et al.*, 2021). It involves conducting research and academic activities in an ethical, consistent and unbiased manner (Guerrero-Dib *et al.*, 2020; Ramdani, 2018). Studies suggest that students often resort to cheating during exams due to the pressure to achieve higher grades. Conversely, academically successful students are generally less inclined to engage in dishonest practices (Lee *et al.*, 2020). Additionally, research shows that students with negative attitudes toward cheating tend to achieve better grades, further highlighting the importance of fostering academic integrity, especially in the context of advanced AI technologies (Vendehey *et al.*, 2007). The relationship between academic integrity and academic performance is widely supported in the literature. Bin-Nashwan *et al.* (2023a) found that academic integrity negatively impacts the adoption of ChatGPT by academicians, underscoring concerns about ethical AI usage. Furthermore, students with a track record of academic success are more likely to trust their ability to succeed, reducing their reliance on dishonest practices (Lee *et al.*, 2020). Similarly, Elsalem *et al.* (2021) demonstrated a significant link between academic integrity and the effort and time invested in exam preparation. Academic integrity also plays a critical role in career readiness. Sotiriadou *et al.* (2020) revealed that promoting academic integrity improves the employability of graduates. However, despite these findings, there remains a gap in understanding the factors shaping academic integrity among academic staff and the

key drivers of this essential practice. This study seeks to address this gap by examining the influences on academic integrity in the evolving landscape of the ChatGPT era.

*2.3.2 Personal best goals.* PBG are defined as challenging, specific, self-improvement-focused and competitively self-referenced objectives that encourage individuals to surpass their previous achievements (Martin and Elliot, 2016). These goals uniquely contribute to explaining variance in academic outcomes, highlighting their critical role in motivation and educational engagement (Benlahcene, 2022; Martin et al., 2022). In a longitudinal cross-lagged panel study conducted in Australia, PBG were shown to influence high school students' implicit beliefs and intelligence traits (Martin, 2015). Furthermore, research comparing PBG with traditional goal types, such as mastery and performance goals, found that PBG significantly enhanced motivation and engagement among Chinese and Australian elementary and secondary school students (Martin and Elliot, 2016). A study involving middle and secondary school students in Australia revealed that students in a PBG-focused setting achieved greater academic growth compared to those in a control group (Martin and Elliot, 2016). Additional research reinforces the positive impact of PBG in higher education contexts. For example, Ramshe et al. (2019) demonstrated a significant positive relationship between PBG and various dimensions of learning engagement – behavioral, emotional and cognitive – among 302 university students. Similarly, Khajavy and Aghaee (2022) found that PBG significantly influenced students' academic performance. Burns et al. (2018) also provided evidence that PBG significantly predicted both student engagement and academic integrity, supporting the integration of PBG into models like the SCT. Recently, McNulty (2023) emphasized the need for further research to fully understand PBG's effects on behavioral improvements in academic practices. Based on this evidence, it is reasonable to assume that PBG are strongly linked to academic integrity, particularly in the context of the ChatGPT era:

*H1.* Personal best goals relate to academic integrity in the ChatGPT era.

*2.3.3 Academic competence.* In academia, competence encompasses a blend of knowledge, attitudes and skills that enable individuals to effectively perform tasks and solve problems (Wiek et al., 2011). Instructors play a crucial role in fostering students' sense of competence by promoting honest communication, providing clear and consistent guidelines and cultivating meaningful interactions between students and faculty (Muñoz and Ramirez, 2015). Bandura's (1986) concept of self-efficacy underscores the importance of perceived competence in influencing behavior (Teoh et al., 2020). Research has highlighted the significant role of competence in shaping academic practices. For example, Liu et al. (2015) conducted a study involving 462 undergraduate students in Hong Kong and demonstrated that competence positively impacts ethical academic activities. Similarly, Panayiotou et al. (2019) found in their experiment with 1,626 students in England that competence significantly affects academic achievement. Manwa et al. (2020) also reported that competence influences the academic practices of female university students in Zimbabwe. However, a systematic review by Luesia et al. (2023) revealed limited evidence regarding the relationship between competence and academic integrity. This gap in the literature highlights the need for further research to explore the connection between competence and ethical academic behaviors, such as integrity, to better understand this critical relationship:

*H2.* Academic competence relates to academic integrity in the ChatGPT era.

*2.3.4 Workplace stress.* Workplace stress, as defined in our research, refers to an individual's subjective experience of pressure or tension over a specific time frame (Fisher, 2015; Cohen et al., 1983). This stress emerges when individuals face persistent inconveniences, challenges or issues that substantially impact critical aspects of their professional or personal lives. Recent research highlights the adverse effects of stress on mental health, particularly within the context of online learning (Heo and Han, 2018). The

abrupt shift to online education during the COVID-19 pandemic has been identified as a significant stressor, exacerbating stress levels among students (Mheidly *et al.*, 2020). Findings in the literature present mixed results regarding the relationship between stress and academic outcomes. For instance, a study involving 214 psychology students in Saudi Arabia found no correlation between perceived stress and academic achievement (Mohsen, 2017), while Tus (2020) reported similar outcomes. Interestingly, perceived stress was also associated with an increase in academic practices – both positive and negative – during the pandemic. However, other studies, such as that of Gustems-Carnicer *et al.* (2019), identified a significant negative relationship between high levels of stress and academic performance. Building on these insights, this study posits that workplace stress negatively impacts academic integrity by impairing self-regulation and ethical decision-making. Elevated stress levels may lead individuals to prioritize short-term relief or outcomes over long-term values, thereby increasing the likelihood of unethical behavior. Stress can further weaken cognitive functioning, reducing the capacity to evaluate ethical considerations effectively. Moreover, it may create a sense of urgency or desperation to meet demands, fostering an environment where academic misconduct becomes more appealing:

H3. Workplace stress relates to academic integrity in the ChatGPT era.

*2.3.5 ChatGPT adoption.* In education, AI adoption has emerged as a consequential tool in individuals' achievement and academic practices, garnering substantial attention within the academic community (Hisan and Amri, 2023; Lo, 2023). This technology demonstrates the capacity to swiftly create text that closely resembles human language in a conversational approach using machine learning techniques. Its remarkable versatility extends to handling diverse language prompts, including translating text, answering questions, summarising information and generating new textual content (Cotton *et al.*, 2024). However, using ChatGPT in academia also poses substantial challenges, including the risk of plagiarism, academic dishonesty and AI-assisted fraud (Shiri, 2023; Cotton *et al.*, 2024; Lo, 2023; Alser and Waisberg, 2023). Also, it questions the extent to which educators develop necessary skills and gain from academic challenges that should ameliorate their readiness for future endeavors. As a result, numerous universities globally have opted to restrict access to ChatGPT on their campuses (Tlili *et al.*, 2023), while others remain cautious due to concerns about disseminating unreliable, unethical or plagiarised content.

The growing intellectual discourse surrounding ChatGPT's impact on academia and higher education highlights its contentious role in shaping academic practices, including ethics and integrity (Lo, 2023; Cotton *et al.*, 2024; Alser and Waisberg, 2023; Farrokhnia *et al.*, 2023; Lund and Wang, 2023; Rahman *et al.*, 2023; Dowling and Lucey, 2023; Dwivedi *et al.*, 2023; Jimenez, 2023). While some researchers, such as Malinka *et al.* (2023), have argued that ChatGPT can significantly enhance academic practices, they have warned that its current use – absent adequate and responsible user training – may undermine academic quality. This divergence in findings underscores the need for further investigation into the impact of ChatGPT adoption on ethical academic practices. Moreover, there is a notable gap in the literature concerning ChatGPT's moderating role in the relationship between academic integrity and its predictors. To address this gap and advance understanding in this area, this study proposes the following hypotheses:

H4. ChatGPT adoption relates to academic integrity in the ChatGPT era.

H4a. ChatGPT adoption moderates the relationship between personal best goals and academic integrity in the ChatGPT era.

H4b. ChatGPT adoption moderates the relationship between competence and academic integrity in the ChatGPT era.

H4c. ChatGPT adoption moderates the relationship between perceived stress and academic integrity in the ChatGPT era.

### 3. Research methods

In this attempt, the objective was to provide a comprehensive understanding of trends and relationships by manipulating a limited number of variables to explore theory-driven research questions and hypotheses (Bin-Nashwan *et al.*, 2023b). A cross-sectional research design was used to validate the academic integrity model through ChatGPT adoption, with a web-based questionnaire distributed to the global academic community. The online survey was chosen for its efficiency, cost-effectiveness and accessibility to the target population (Wright, 2005), which aligns with the technological nature of this study. The respondents were academic scholars worldwide, particularly those active on Academia.edu and ResearchGate, with at least two identifiable research publications on their profiles. These platforms are among the most widely used academic social networking sites (ASNSs) worldwide (Bin-Nashwan *et al.*, 2023a, 2023b), with ResearchGate boasting 20 million users and Academia.edu reaching 170 million users in 2021 (ResearchGate, 2021; Academia.edu, 2021). For data collection, survey invitations containing URL links were randomly sent to respondents on RG and Academia.

Prior to distributing the survey, we sought input from three academics and conducted a pilot test with a group of 20 researchers. Based on their feedback, several refinements were made to enhance the validity and reliability of our proposed model (Abdul-Jabbar and Bin-Nashwan, 2022). To minimize potential social desirability bias, we implemented a series of careful measures, including ensuring the confidentiality of responses, conducting an extensive pilot study and adopting a web-based survey design (Dodou and de Winter, 2014). The sample revealed a notable gender disparity among the academic staff and researchers, with 68.2% identifying as male and 31.8% as female. Regarding age, 39.6% of the respondents were between the ages of 31 and 40, while 30.6% were aged between 41 and 50. When it came to academic roles, 26.6% were Senior Lecturers/Assistant Professors, 24.4% were Instructors/Lecturers, 20% were Postdoctoral Researchers/Fellows/Scholars, 17% were Associate Professors and 12% held the title of Professor. Furthermore, participants' academic experience varied, with 32.6% reporting 1–5 years of experience, and 28.1% having 11–15 years of experience in academia. These insights provide a comprehensive understanding of the academic backgrounds and demographic distribution of the study sample, helping to contextualize the results.

To ensure the validity of the study model, it was essential to assess the potential impact of common method bias (CMB, Bin-Nashwan *et al.*, 2022). Therefore, Harman's single-factor analysis was conducted to determine whether a single construct could explain a substantial portion of the variance in the proposed model. The results showed that the largest single factor explained only 27.50% of the variance, which is well below the recommended threshold of 50%, as suggested by Podsakoff *et al.* (2003). This indicates that CMB was not a significant issue, confirming the validity of the study's findings. Additionally, a thorough collinearity test was conducted to examine potential collinearity problems within the model. The variance inflation factor values were all below the critical threshold of 3.3 (Bin-Nashwan, 2022), which further supports the model's validity and the reliability of the results.

We relied upon an online survey for assessing five variables: PBG, academic competence, workplace stress, ChatGPT adoption and academic integrity in the ChatGPT era. Each measurement item was derived from well-established instruments used in prior research. These items have been carefully modified to match the specific context of academic performance in the ChatGPT era. PBG were measured using three questions adapted from Martin (2006), while competence was evaluated using five items adapted from Chen *et al.* (2015). Six items drawn from Cohen *et al.* (1983) were used to measure stress, and five items derived from Katz *et al.* (1973) and Papacharissi and Rubin (2000) were used to evaluate ChatGPT adoption in academia. Academic integrity was measured via 17 items derived from Ramdani (2018). A five-point Likert-type scale, ranging from 1 "strongly disagree" to 5 "strongly agree", was used to evaluate the perceptions of academicians who

participated in this study. As for data analysis software, we used partial least squares-structural-equation-modeling – a robust analytical tool widely used to explore hypothesized relationships among variables (Sadallah *et al.*, 2023) – to analyze the data collected. As a variance-based modeling tool, this software has gained considerable interest from researchers from various fields, including social sciences and management. It has the potential to deal with issues, e.g. non-normal data distributions, limited sample sizes and complexity of the relationships between the underlying latent variables (Hair *et al.*, 2017).

## 4. Findings

### 4.1 Measurement model

An in-depth evaluation of the validity and reliability of the variables was conducted before testing the structural model to assess the reflective model in this study. This process is crucial for confirming internal consistency, as well as both convergent and discriminant validity, and ensuring the reliability of the constructs. Hulland (1999) recommended a minimum threshold of 0.4 for item loadings, while Hair *et al.* (2017) argued that a loading above 0.70 is necessary to achieve adequate validity and reliability. However, Hair *et al.* (2017) also suggested that items with loadings between 0.40 and 0.70 could be removed if doing so improves composite reliability (CR). Accordingly, five items – AI3, AI6, AI7 and AI9 – were excluded from the original survey. The convergent validity is established as the values of the average variance extracted (AVE) exceeded the cut-off criterion of 0.50, as Hair *et al.* (2017) advocated. The internal consistency of the measures was assessed by all latent CR values in Table 1, exceeding the required threshold of 0.70, ensuring the constructs' reliability. It is critical for assuring the validity and reliability of every variable to prevent shortcomings in study findings and drawn conclusions.

| Table 1 Measurement model results |      |         |       |       |
|-----------------------------------|------|---------|-------|-------|
| Construct                         | Item | Loading | CR    | AVE   |
| Personal best goals (PBG)         | PBG1 | 0.825   | 0.914 | 0.787 |
|                                   | PBG2 | 0.911   |       |       |
|                                   | PBG3 | 0.815   |       |       |
| Academic integrity (AI)           | AI1  | 0.824   | 0.899 | 0.592 |
|                                   | AI2  | 0.840   |       |       |
|                                   | AI4  | 0.786   |       |       |
|                                   | AI5  | 0.744   |       |       |
|                                   | AI8  | 0.817   |       |       |
|                                   | AI10 | 0.821   |       |       |
|                                   | AI11 | 0.873   |       |       |
| Academic competence (AC)          | AI12 | 0.811   | 0.889 | 0.614 |
|                                   | AC1  | 0.823   |       |       |
|                                   | AC2  | 0.812   |       |       |
|                                   | AC3  | 0.766   |       |       |
|                                   | AC4  | 0.751   |       |       |
| Workplace stress (WS)             | AC5  | 0.817   | 0.938 | 0.799 |
|                                   | WS1  | 0.832   |       |       |
|                                   | WS2  | 0.912   |       |       |
|                                   | WS3  | 0.921   |       |       |
|                                   | WS4  | 0.893   |       |       |
|                                   | WS5  | 0.762   |       |       |
| ChatGPT adoption (GPT)            | WS6  | 0.891   | 0.919 | 0.691 |
|                                   | GPT1 | 0.941   |       |       |
|                                   | GPT2 | 0.863   |       |       |
|                                   | GPT3 | 0.823   |       |       |
|                                   | GPT4 | 0.783   |       |       |
|                                   | GPT5 | 0.715   |       |       |
| Source(s): Authors' own work      |      |         |       |       |

The evaluation of the measurement model includes an important aspect known as discriminant validity, which was assessed using the heterotrait-monotrait ratio approach. This method measures the correlation coefficients between two items belonging to the same variable. The results, presented in Table 2, show that the constructs are adequately distinguished from each other, as all construct ratios are below the recommended threshold of 0.85 (Kline, 2023). Furthermore, to evaluate the reliability and validity of the measurement model, several statistical analyses were conducted. The results confirm that the measurement instruments used to assess the constructs in this study demonstrate both reliability and validity, thereby allowing for the assessment of the proposed structural equation model. These tests confirm the robustness of the measurement model and its ability to accurately measure the underlying constructs of interest.

4.2 Structural model

The estimate of the structural model primarily focuses on analyzing the relationships between the exogenous and endogenous constructs. This assessment includes examining collinearity among the constructs and determining the significance of the hypothesized relationships (Hair et al., 2016). To assess the significance of these relationships, a bootstrapping technique with 5,000 samples was used. The statistical analysis results showed that all path relationships were statistically significant, as evidenced by bootstrap critical *t*-values exceeding ±1.65 (one-tailed test). This supports the presence of meaningful and influential connections between the exogenous and endogenous constructs.

Our data analysis outcomes further highlight significant relationships between several constructs and academic integrity in the ChatGPT era. Specifically, PBG ( $\beta = 0.129, p = 0.003$ ), and ChatGPT use ( $\beta = 0.093, p = 0.009$ ) were positively associated with academic integrity. However, workplace stress ( $\beta = -0.254, p = 0.000$ ) was found to have a negative relationship with academic integrity. Whereas academic competence ( $\beta = 0.033, p = 0.223$ ) had no effect on academic integrity. Hence, these results statistically support hypotheses *H1, H3 and H4*, but did not support *H2*.

The results regarding the interaction relationships are presented in Table 3. The interaction effects between PBG × ChatGPT adoption ( $\beta = -0.077, p = 0.041$ ), academic competence × ChatGPT adoption ( $\beta = 0.299, p = 0.000$ ) and workplace stress × ChatGPT adoption ( $\beta = -0.134, p = 0.000$ ) on academic integrity in the ChatGPT era were found to be significant. These findings provide support for all the interaction hypotheses, namely *H4a, H4b and H5c*.

5. Discussion

In the rapidly evolving academic landscape, the ChatGPT era marks a transformative phase where human intellect converges with the innovative capabilities of AI. This integration has

| Table 2 Discriminant validity                                                                                                                        |       |       |       |       |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|---|
| HTMT 0.85 criterion                                                                                                                                  | 1     | 2     | 3     | 4     | 5 |
| 1 AI                                                                                                                                                 |       |       |       |       |   |
| 2 PBG                                                                                                                                                | 0.049 |       |       |       |   |
| 3 AC                                                                                                                                                 | 0.082 | 0.061 |       |       |   |
| 4 WS                                                                                                                                                 | 0.155 | 0.341 | 0.051 |       |   |
| 5 GPT                                                                                                                                                | 0.082 | 0.418 | 0.242 | 0.393 |   |
| Note(s): AI = academic integrity, AC = academic competence, PBG = personal best goals, WS = workplace stress; ChatGPT = ChatGPT adoption in academia |       |       |       |       |   |
| Source(s): Authors' own work                                                                                                                         |       |       |       |       |   |

**Table 3** Results of inner (structural) model

| Hypotheses | Path                          | $\beta$ | Std error | t-value | p-value | Supported? |
|------------|-------------------------------|---------|-----------|---------|---------|------------|
| H1         | PBG $\rightarrow$ AI          | 0.129   | 0.048     | 2.675   | 0.003   | Yes        |
| H2         | AC $\rightarrow$ AI           | 0.033   | 0.053     | 0.663   | 0.223   | No         |
| H3         | WS $\rightarrow$ AI           | -0.254  | 0.066     | 3.848   | 0.000   | Yes        |
| H4         | ChatGPT $\rightarrow$ AI      | 0.093   | 0.042     | 2.214   | 0.009   | Yes        |
| H4a        | PBG* ChatGPT $\rightarrow$ AI | -0.077  | 0.040     | 1.925   | 0.041   | Yes        |
| H4b        | AC* ChatGPT $\rightarrow$ AI  | 0.299   | 0.083     | 3.602   | 0.000   | Yes        |
| H4c        | WS* ChatGPT $\rightarrow$ AI  | -0.134  | 0.043     | 3.113   | 0.000   | Yes        |

**Note(s):** AI = academic integrity, AC = academic competence, PBG = personal best goals, WS = workplace stress; ChatGPT = ChatGPT adoption in academia

**Source(s):** Authors' own work

significantly reshaped the dynamics of academic integrity among faculty, scholars and researchers. Traditional measures of academic success now intersect with the groundbreaking potential of AI-driven tools like ChatGPT, redefining the core principles and evaluation of academic practices. This study aims to explore the complexities of academic integrity in the context of the ChatGPT era, focusing on key predictors such as PBG, academic competence and workplace stress. Additionally, it examines the moderating role of ChatGPT adoption, shedding light on how these factors collectively influence the ethical conduct of academic professionals in today's scholarly environment.

In this research, we found a positive relationship between PBG and academic integrity among academic staff. This can be attributed to the intrinsic motivation and self-regulation that PBG foster. According to SCT, individuals are motivated by self-regulation and self-efficacy (Bandura, 1986). That is, when academic staff strives to outperform their own previous achievements rather than competing with others, they are more likely to prioritize ethical behavior and uphold integrity in their academic endeavors. This self-referenced standard encourages accountability and a commitment to authentic success, aligning with the principles of academic integrity. Consequently, PBG act as a driving force that reinforces ethical conduct by emphasizing personal growth and professional excellence. However, we found that academic competence had no direct effect on integrity in the ChatGPT era. This suggests that possessing knowledge and skills alone may not necessarily translate to ethical behavior in academia. This could be due to the increasing reliance on AI tools like ChatGPT, which may shift the focus from individual competence to collaborative or technology-assisted outputs. Furthermore, integrity might be more influenced by factors such as personal values, external pressures or institutional culture rather than by an individual's level of expertise. This indicates that ethical behavior in the ChatGPT era is shaped by a broader set of social and contextual factors beyond just academic competence. While competence is a core factor in SCT (Bandura, 1986), the absence of a direct relationship with integrity suggests that other external factors, such as social environment or the availability of AI tools, might play a more significant role in influencing behavior. This aligns with SCT's emphasis on the impact of external environments in shaping individuals' actions and decisions (Bandura, 1989).

The study also reported a negative relationship between workplace stress and academic integrity while using ChatGPT. The negative relationship between workplace stress and academic integrity, particularly in the context of ChatGPT usage, is supported by SCT's assertion that stress can diminish self-regulation and decision-making capacity (Bandura, 1989). High levels of stress may impair an individual's ability to maintain ethical standards, leading to a greater likelihood of unethical academic practices when using AI tools. This result highlights how stress can undermine ethical decision-making in academia. High levels of stress may push individuals toward shortcuts or reliance on AI tools like ChatGPT in ways that compromise integrity, such as bypassing original work or failing to adhere to

ethical standards. Stress can also impair judgment, reduce self-regulation and heighten the likelihood of unintentional errors or ethical lapses. In the ChatGPT era, the accessibility of AI tools might exacerbate this issue, as stressed academics may over-rely on such tools without critically evaluating their use. However, the research revealed that academicians' integrity in the ChatGPT era was positively related to their perceived ChatGPT adoption. This discovery suggests that the tool can support ethical practices in academia when used responsibly. By providing quick access to reliable information, aiding in research processes and streamlining administrative tasks, ChatGPT enables academics to focus on higher-order activities, reducing the likelihood of errors or unethical shortcuts. Additionally, its potential for transparency and accuracy in generating content can encourage proper citation and adherence to ethical standards. This finding implies that when integrated thoughtfully, ChatGPT can enhance the overall integrity of academic practices.

By integrating ChatGPT usage as a moderating factor in the relationships between academic integrity and its key determinants – PBG, academic competence and workplace stress – we discovered some intriguing findings. These findings can be explained by SCT's focus on how environmental factors (in this case, AI tools) influence behavior. The analysis revealed that ChatGPT usage in academia had a negative moderating effect on the previously established positive relationship between PBG and academic integrity. This suggests that while striving for personal achievements typically fosters ethical behavior, the reliance on ChatGPT may alter this dynamic. The ease and efficiency of AI tools might inadvertently shift the focus from genuine effort and self-improvement to task completion through automation. This could diminish the intrinsic motivation tied to PBG, potentially leading to a weakened commitment to ethical standards. As a result, the use of ChatGPT may dilute the self-driven ethos that traditionally underpins academic integrity.

The analysis further revealed that ChatGPT adoption in academia positively moderated the relationship between academic competence and academic integrity. This suggests that AI tools like ChatGPT can enhance the ethical standards of academically competent individuals. For scholars with strong academic skills, ChatGPT provides an avenue to refine their work, access reliable information and manage complex tasks efficiently, all while upholding high ethical standards. When used appropriately, this technology can act as a supportive resource, helping competent academics maintain both efficiency and integrity by reducing the temptation to cut corners. Additionally, the study found that ChatGPT usage in academia amplified the negative relationship between workplace stress and academic integrity. This finding indicates that increased reliance on AI tools may exacerbate the ethical challenges associated with stress. While ChatGPT can reduce workloads and streamline tasks, it may also lead to stress-induced over-reliance, resulting in ethical lapses such as improper citation or producing unoriginal work. In high-stress environments, the convenience of ChatGPT could tempt academic staff to prioritize expediency over integrity, thereby intensifying the adverse effects of workplace stress on ethical behavior.

## 6. Implications and conclusions

This study provides valuable insights into the integration of AI-driven chatbots like ChatGPT in academic environments. Through empirical research, it explores a relatively new area, contributing to existing scholarly works by examining key factors that affect academic performance through the use of this innovative tool. Additionally, it highlights the important role of academic integrity among faculty and researchers in an AI-driven era. These findings establish a solid foundation for future research, offering meaningful implications that could drive further progress in understanding the impact of AI technology on academic practices.

First, due to the relevance of this study's topic and the clear theoretical gaps in existing literature, particularly concerning the adoption of ChatGPT in academic settings, this

research stands as one of the first empirical investigations into the use of OpenAI among academic staff within educational environments. Second, building on insights from SCT, this study makes a valuable contribution to the literature by examining the global use of ChatGPT among academics, focusing on the most widely used ASNSs like ResearchGate and Academia. Third, the model developed in this study integrates several key factors – PBG, competence and workplace stress – that may influence academic integrity when using ChatGPT. These factors have been underexplored in AI-driven educational contexts, highlighting a significant gap in empirical research. Therefore, the model established here provides a solid empirical foundation for future studies in AI applications, offering a path for further exploration of this critical innovation. Additionally, our study provides important and timely evidence on the use of ChatGPT in academic environments, revealing significant moderating effects on the academic integrity of faculty and researchers.

Practically, the results of this study provide actionable managerial insights and suggestions for academia in the ChatGPT era. Specifically, various stakeholders, including educational institutions, faculty, researchers and ethics and research committees, can use these findings to inform their strategies and decisions. Stakeholders like educational institutions must prioritize comprehensive ethics education programs for faculty members and researchers that specifically address the ethical utilization of AI tools such as ChatGPT. This will instill a robust foundation of academic integrity within technological advancements. Clear protocols and guidelines must be established for the ethical deployment of AI in research and academic endeavors. Institutions should encourage academic staff to set PBG, which can positively influence their academic integrity. However, it's crucial to provide support and resources that reduce stress, ensuring that AI tools do not diminish the intrinsic motivation to achieve ethical, high-quality work. The negative relationship between workplace stress and academic integrity suggests a need for institutions to foster a supportive work environment. Offering stress management programs, workload balancing and mental health support can help reduce stress and promote ethical behavior among academic staff. Furthermore, research and ethics committees should actively monitor and assess the use of AI tools like ChatGPT in academic research to ensure that AI-assisted work adheres to ethical standards. They should also promote ongoing discussions about the potential ethical challenges posed by AI and foster a culture of transparency in its use. ChatGPT's positive moderating effect on the relationship between academic competence and integrity suggests that integrating AI can enhance academic staff's performance and integrity. Training programs should be designed to help faculty and researchers leverage AI to enhance their academic competence while maintaining high ethical standards. AI has the potential to provide tailored learning experiences by adapting to the individual needs of students, offering personalized feedback and facilitating more efficient administrative processes. Educators could leverage these tools to streamline content delivery, assess students' understanding in real-time, and create more dynamic, interactive learning environments. Additionally, the use of AI can help reduce the burden of repetitive tasks, allowing educators to focus more on meaningful interactions with students. However, it is essential to ensure that AI adoption is coupled with ethical considerations to maintain academic integrity and support the development of critical thinking skills. Further research into the best practices for integrating these tools can help guide educators in maximizing their potential while preserving the quality and authenticity of education.

This study used SCT to examine academic integrity in the context of ChatGPT adoption within educational settings and academia. Positioned as a foundational exploration, future research could broaden the theoretical lens by incorporating additional relevant frameworks, such as expectancy-value theory, interest theory and achievement goal theory. These perspectives could provide a more nuanced understanding of technology use in academia, offering deeper insights into this critical topic. While ResearchGate and Academia.edu were selected as the primary ASNSs due to their prominence, they may not fully represent all users across diverse ASNSs. Future studies could investigate other significant platforms, such as Google Scholar, LinkedIn and Mendeley, to present a more

holistic perspective on ChatGPT adoption within academic communities. Moreover, expanding the scope to include students and other academic staff could shed light on their motivations for engaging in positive academic practices with AI tools, further enriching the understanding of the impact of such technologies on various academic stakeholders.

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## Further reading

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